Qns	Answer	Marks
2(a)(i)	An arrow originating from the object, pointing towards point O, and labelled Z.	B1
2(a)(ii)	The frictional force provides the centripetal force, and hence must point towards the centre of the disc.	B1
2(b)(i)	Using points (6.00, 5.90) and (9.00, 8.90) on the line $\text{Gradient} = \frac{8.90 - 5.90}{9.00 - 6.00} = 1.00$	B1 B1
2(b)(ii)	$\text{gradient} = \frac{\omega_{\max}^2}{\frac{1}{r}} = r\omega_{\max}^2 = a_{c(\max)}$ Hence, the gradient is numerically equal to the **maximum** centripetal acceleration.	B1 A1*

2(c)	$f = \text{gradient} \times m$ $= 1.00 \times 0.80$ $= 0.80 \text{ N}$	C1 A1
(d)	As angular speed increases, **required** centripetal force increases. (friction provides required centripetal force) When required centripetal force exceeds maximum frictional force possible, object slides.	B1 B1

Qns	Answer	Marks
3(a)(i)	**Free oscillations** are oscillations with	